

Weak Prompts

Recommend me some movies.

I want a good movie.

What should I watch tonight?

Give me Netflix movies.

Tell me the best movie ever

Strong Prompts

Recommend 5 science fiction movies similar to *Interstellar*. I like movies with space exploration, emotional storytelling, and complex ideas. Include a short explanation for why I might enjoy each movie.

I want a comedy movie to watch with my family. Avoid anything rated R and recommend movies that are funny for both teenagers and adults.

Recommend horror movies that are scary but do not rely heavily on gore. I prefer psychological horror and suspense.

I enjoyed *The Dark Knight*, *Inception*, and *Blade Runner 2049*. Recommend movies with similar themes, directing styles, or storytelling.

I only have 90 minutes tonight. Recommend highly-rated movies under 1 hour and 45 minutes.

Hallucination Prompts

Tell me about the movie *The Last Ocean Planet* (2027) starring Chris Evans.

Recommend movies starring Emily Watson Jr., especially her superhero films.

Current Information Test

Is *Avatar 5* available on Netflix right now?

Recommend movies directed by Christopher Nolan that were released before 1900.

Fake Award Test

Explain why Titanic 2 won Best Picture at the Oscars.

False Test

Why was Finding Nemo judged for having too many musical numbers?

Conversation-Adaptive Recommendation Test

I don't know what I like. Ask me questions and help me discover what type of movies I would enjoy.

We've discussed the importance of prompt engineering in this class and wanted to test our AI using weak prompts, strong prompts and hallucination prompts mainly. Weak prompts test how the AI deals with missing information. Strong prompts test accuracy and personalization. Hallucination prompts tests whether the AI admits uncertainty instead of generating false movie information.

For testing our movie recommendation AI, we divided our prompts into several categories. First, we used weak prompts, such as 'Recommend me some movies,' to see if the AI recognizes when it doesn't have enough information and asks the user follow-up questions.

Next, we used strong prompts with specific preferences, such as genre, favorite movies, audience, and runtime. These test whether the AI can actually personalize its recommendations and follow multiple constraints.

We also created hallucination tests using fake movies, fake actors, false awards, and impossible scenarios. The goal here is to make sure the AI doesn't simply make up information when the user gives it a false premise. A correct response should acknowledge that the information is false or cannot be verified.

Finally, we tested more interactive situations, such as asking the AI to interview a user who doesn't know what they want to watch. This tests whether the AI can ask useful questions and adapt its recommendations based on the conversation.

Overall, we're testing not only whether our AI can recommend movies, but whether it can understand the user's intent, follow their preferences, ask for clarification when necessary, and avoid hallucinating information.

Lessons Learned

Through testing, we learned that the quality and specificity of a prompt can have a major impact on the quality of an AI's response. We also learned that testing an AI isn't just about whether it gives a good recommendation, but whether it can handle missing information and avoid making up information when given a false question.